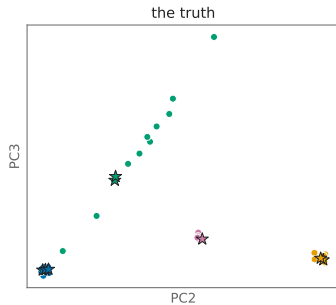
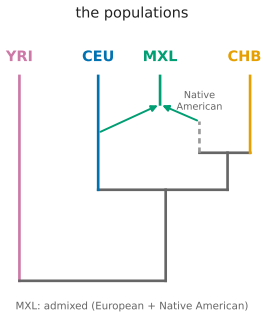
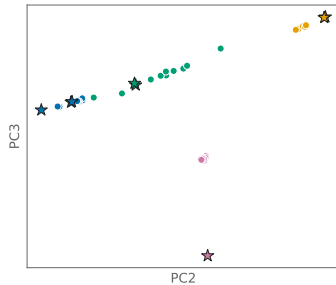


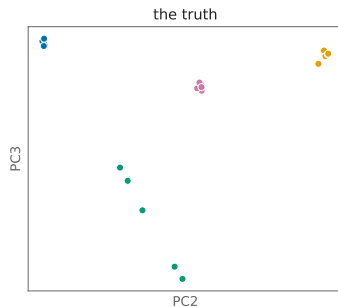
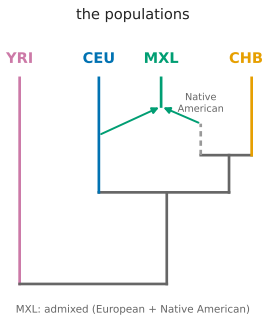


● CEU ● CHB ● MXL ● YRI ☆ relative

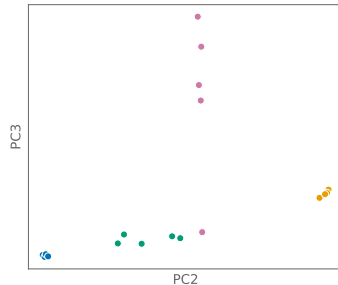
standard PCA, 10 per population + 8 relatives  
PC3 recovered:  $R^2 = 0.01$



**8 relatives among 40 people — and one ancestry axis is gone.**



standard PCA, 5 per population, no relatives  
PC3 recovered:  $R^2 = 0.02$



5 people per population, no relatives — and PCA fails again.

# Relatedness in PCA

and how to deal with it

relatePCA

October 5, 2026

# The goal

- PCs that reflect **ancestry**, not families.
- **Everyone in-sample**: nothing dropped, nothing projected.
- From **20 to 100,000** individuals.
- And estimate the relatedness itself — also for **admixed** people.

# Why relatives break PCA

# Relatives share alleles

- PCA looks at how genotypes co-vary between individuals.
- Relatives co-vary beyond what ancestry explains:

$$\text{Cov}(g_i, g_j) = 4 \phi_{ij} \pi(1 - \pi)$$

- A family is a small, tight cluster — PCA spends components on it.

## Large samples: a different symptom

| $N = 2000$ , 30% in families   | standard PCA | robust |
|--------------------------------|--------------|--------|
| ancestry axes correct          | yes          | yes    |
| relatives' placement error     | 0.49         | 0.12   |
| family axes among 16 extra PCs | 16           | 0      |

**Small N: a lost axis. Large N: misplaced relatives, family axes.**

## The usual fixes

- **Drop relatives** — you lose people.
- **Project relatives** (PC-AiR) — they are no longer in-sample.
- **Screen with KING** — biased for admixed people (later).

**We want to keep everyone in the PCA.**



# The small- $N$ problem

# The GRM diagonal

- Diagonal = structure + heterozygosity noise.
- At 5 per population the noise wins — and it differs by population.
- PCA spends an axis on it, **even without relatives**.

That was the second hook.

## Fix: ignore the diagonal

- Treat the diagonal as **unobserved**.
- Impute it from the off-diagonal entries (`--impute-diag`).

| 5 per population, mean min $R^2$ | no relatives | with relatives |
|----------------------------------|--------------|----------------|
| diagonal kept                    | 0.02         | 0.02           |
| diagonal imputed                 | <b>0.99</b>  | <b>0.99</b>    |

## A trap: centring

- The GRM uses *centred* genotypes.
- Centring spreads each person's diagonal over the whole matrix:

$$C_{ij} \ni -\frac{d_i + d_j}{N}$$

- Freeing the diagonal then frees nothing.

## Work uncentred

|                             |     |                            |
|-----------------------------|-----|----------------------------|
| low-rank fit, diagonal free |     | failures, 5 per population |
| centred GRM                 | 82% |                            |
| uncentred GRM               | 0%  |                            |

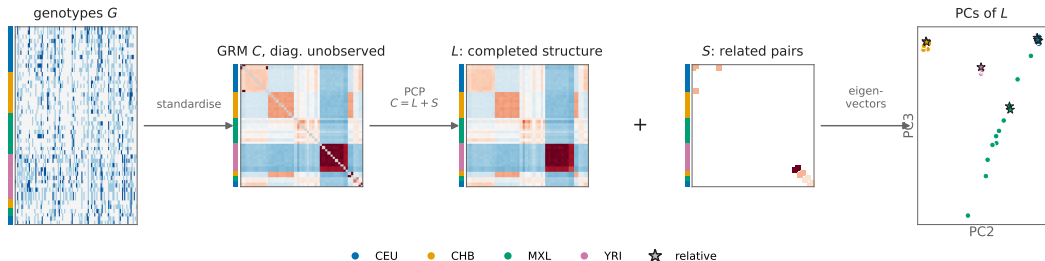
**Scaling does not matter. Centring does.**

# The methods

# The idea

- Split the matrix: **structure** (low rank) + **related pairs** (sparse).
- A pair is related if its kinship  $> \tau = 2^{-3.5} \approx 0.088$  (between 2nd and 3rd degree).
- The diagonal stays free.

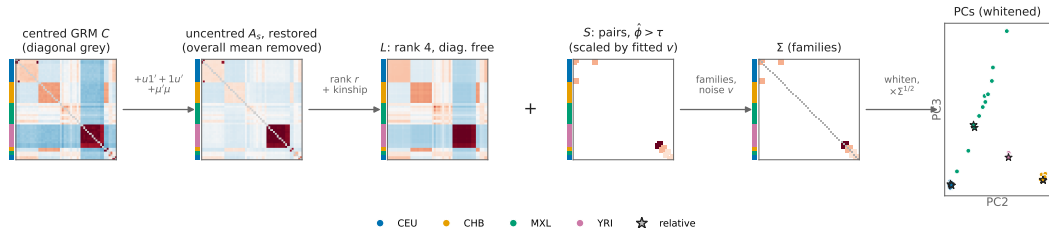
# Robust PCA of the GRM (aarobust-kin)



**Accurate, but slow: a full eigendecomposition every iteration.**



# Detect, then whiten ( $d_{wg}$ , the default)



**Fast: only the top few eigenvectors. No HWE assumption.**

## How many PCs? The data decide

- Add axes while they rise above the **noise edge**.
- An axis resting on 2–3 people is a **family**, not ancestry.
- So  $-k$  is only an upper bound.

| runs that change between $k = 3$ and $k = 10$ | old | dwg      |
|-----------------------------------------------|-----|----------|
| of 240                                        | 12  | <b>2</b> |

## Whiten — don't just remove the pairs

- Removing the related entries still counts each sibling fully.
- Whitening counts a family by its *effective* size.

| family signal in the extra PCs | median $\eta^2$ |
|--------------------------------|-----------------|
| standard PCA                   | 0.92            |
| related pairs imputed          | 0.37            |
| <b>whitening</b>               | <b>0.21</b>     |

# Kinship for admixed individuals

## KING is not structure-free

- Correct only for two people from the **same unadmixed** population.
- Relatives of different ancestry are pulled **down**.

|                                                 |        |
|-------------------------------------------------|--------|
| true 2nd-degree pair (kinship 0.125)            | KING   |
| YRI grandparent – $\frac{1}{4}$ -YRI grandchild | –0.013 |
| CHB aunt – CHB/YRI nephew                       | 0.017  |

**Both are missed by a KING screen.**

## Use individual allele frequencies

- Predict each genotype from the person's own ancestry:  $\pi_{is}$ .
- What is left over is relatedness.
- PC-Relate, PCAngsd and evalAdmix: **one estimator**.

Use  $K - 1$  PCs. More PCs absorb the relatives.

## The same pairs, with ancestry taken into account

| true 2nd-degree pair (0.125)                    | KING   | evalAdmix-type |
|-------------------------------------------------|--------|----------------|
| YRI grandparent – $\frac{1}{4}$ -YRI grandchild | −0.013 | 0.085          |
| CHB aunt – CHB/YRI nephew                       | 0.017  | 0.089          |

## The dwg screen

1. First fit with KING.
2. evalAdmix kinship from the **ancestry axes only**.
3. Confirm with **IBD**: unrelated  $k_0 \approx 1$ , 2nd degree  $\approx 0.5$ .

| relatives found     | KING screen | dwg screen     |
|---------------------|-------------|----------------|
| 5–40 per population | 86–90%      | <b>95–100%</b> |



## Not just kinship: $(k_0, k_1, k_2)$

- Parent–offspring and full sibs both have kinship  $\frac{1}{4}$ .
- $k_0$  tells them apart: 0 vs  $\frac{1}{4}$ .

| estimated, $N = 2000$ | $k_0$ | $k_1$ | $k_2$ |
|-----------------------|-------|-------|-------|
| parent–offspring      | 0.00  | 1.00  | 0.00  |
| full sibs             | 0.25  | 0.50  | 0.25  |
| 2nd degree            | 0.50  | 0.49  | 0.01  |

## Two details that matter

- Keep each person's own frequencies,  $\pi_i \neq \pi_j$ .  
Pooling them calls a parent–offspring pair 83% unrelated.
- Leave the pair's **whole family** out of the frequency fit.

**Does it work?**

## Realistic LD: HAPNEST, $N = 5000$

|                             | standard | detect-white | dwg          |
|-----------------------------|----------|--------------|--------------|
| relatives' placement error  | 0.048    | 0.028        | <b>0.019</b> |
| relatives found             | –        | 97%          | <b>98%</b>   |
| cross-ancestry grandparents | –        | 66%          | <b>87%</b>   |
| time                        | 14 s     | 100 s        | 21 s         |

The missed pairs share less than expected: realised kinship  $< \tau$ .

## Real panel: 1000 Genomes with ASW

| 5–20 per population | PCA wrong | relatives found |
|---------------------|-----------|-----------------|
| standard PCA        | up to 27% | –               |
| aarobust-kin        | 0%        | 78–84%          |
| dwg                 | <b>0%</b> | <b>90–96%</b>   |

**Bonus: it found real relatives in 1000 Genomes.**

# Scaling up

## Large $N$ : no $N \times N$ matrix

| 16 threads, in memory | standard PCA | dwg    |
|-----------------------|--------------|--------|
| $N = 5000$            | 3.5 s        | 4.7 s  |
| $N = 20,000$          | 24 s         | 26.5 s |

**Robust PCA at about the cost of standard PCA.**

## Very large $N$ : find relatives with a sketch

- Comparing all pairs is  $N^2$  — too slow at biobank scale.
- Sketch each genome; check only each person's nearest neighbours.

| $N = 100,000$          | all pairs | sketch        |
|------------------------|-----------|---------------|
| time to find relatives | 162 s     | <b>52 s</b>   |
| related pairs found    | 27,880    | <b>27,880</b> |



## Genotype likelihoods (-G)

- Low depth shrinks genotypes towards the mean.
- Undo the shrinkage; the extra noise lands on the free diagonal.

| PCA wrong at 4×             | 10 per pop. | 5 per pop. |
|-----------------------------|-------------|------------|
| standard PCA on likelihoods | 100%        | 100%       |
| dwg                         | <b>0%</b>   | 12.5%      |

## Try it

```
PCAone -b data -k 10 --robust          # any N; -k is an upper bound  
PCAone -G data.beagle.gz -k 5 --robust # genotype likelihoods
```

- `.eigvecs`: the robust PCs.
- `.relpairs`: related pairs with kinship and  $(k_0, k_1, k_2)$ .

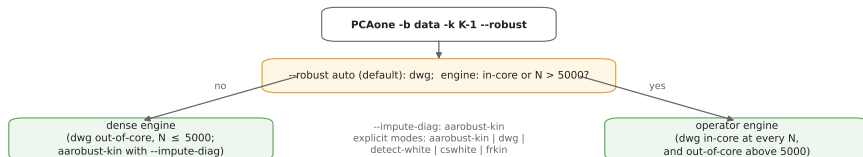
[github.com/aalbrechtsen/PCAone](https://github.com/aalbrechtsen/PCAone), branch `robust-relatives`, docs/`robust/`

## Take home

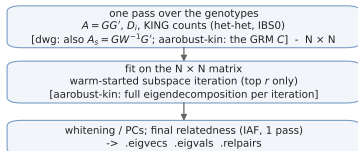
1. Relatives break PCA: a lost axis at small  $N$ , family axes at large  $N$ .
2. At small  $N$  the diagonal breaks it too: work uncentred, diagonal free.
3. Detect the related pairs, then whiten: everyone stays in-sample.
4. For admixed relatives use ancestry-aware kinship, not KING.

Open: missing genotypes; first cousins at  $\sim 5$  per population.

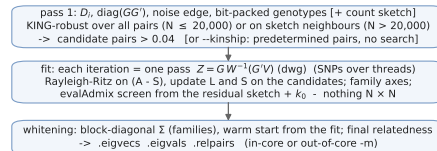
# Backup: the engines



**dense engine (aarobust-kin always; dwg out-of-core and others for  $N \leq 5000$ )**



**operator engine (dwg in-core; other modes  $N > 5000$ )**



## Backup: formulas

- Kinship from the residual:  $\hat{\phi}_{ij} = R_{ij}/(2\sqrt{v_i v_j})$ ,  $R = A - L$ ,  $v_i$  the fitted noise.
- Whitening:  $\Sigma = v^{1/2}(I + 2\Psi)v^{1/2}$ ; PCs of  $\Sigma^{-1/2}A\Sigma^{-1/2}$ , mapped back by  $\Sigma^{1/2}$ .
- Centring:  $JDJ = D - (d\mathbf{1}' + \mathbf{1}d')/N + \bar{d}\mathbf{1}\mathbf{1}'/N$ .
- IBD likelihood: IBD0  $g_i \sim \text{HWE}(\pi_i)$ ,  $g_j \sim \text{HWE}(\pi_j)$ ; IBD1 one shared allele; IBD2  $g_i = g_j$ .